An improved smartphone app to teach transfer function concepts

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***Abstract*— In this paper we present a new version of a smartphone app that serves as a learning tool for the analysis of linear systems modelled as transfer functions. In particular, given a system, the app provides its step response, its pole-zero map, and its Bode and Nyquist plots. The tool is interactive, so that the student can perform a "what-if" analysis when a parameter is changed and understand the relationships between the parameters and the time and frequency responses. Illustrative examples are given to show the main features and how it can be used in control courses.**

I. INTRODUCTION

Especially after the covid-19 pandemics, it has been recognized that it is really important to replace traditional teaching methods (i.e., standard frontal lectures) by new ones that exploit software and hardware tools to provide a more personalized control education [23], [25]. In fact, in addition to the transmission of the control concepts, teaching should also aim to develop in the students those soft skills (such as problem solving) that are essential to manage the today's more ad more complex systems [22].

In this context, it is useful to provide students with smartphone apps that they can use to better understand concepts by visualization. The clear advantage of using a smartphone with respect to a standard laptop is that it can be used more easily in many situations (thus helping in engaging the students more) and it is also more practical to be employed in classrooms so that a given concept explained by the teacher can be immediately verified and therefore better understood. In order to be as fruitful as possible, it is important to provide these apps with a high level of interactivity. Indeed, it has been already largely discussed in the control education community that interactivity is very effective in learning many concepts. Actually, performing a "what-if" analysis allows the student to fully understand the physical meaning of a parameter because what happens when it is changed can be immediately visualized [5], [11]. For this reason, many interactive tools have been created and presented in the last years for many specific topics such as PID control [2], [7], [13], [14], dead time compensation [19], feedforward control [12], linearization [9], fractional control [6], Nyquist criterion [3] and system identification [1], [20], [16], [17], [15]. Such a kind of tools have been developed using various software, including Matlab/Simulink [18], Modelica [26], Easy Java Simulations [24], and Sysquake [8], [10], which are thought to be used with a personal computer.

Motivated by the need of providing an interactive smartphone app, in a previous paper we have presented a tool that plots step responses and Bode plots [4]. In this paper we present a significantly improved version of this app by including the pole-zero map and the Nyquist plot. In this way, the student can perform a complete analysis of a transfer function, which is one of the most relevant topic to be addressed in a first control course [21]. In particular, the student can analyze first- and second-order transfer functions (either with complex conjugate or real poles and with the optional addition of a real zero) by analyzing interactively what happens by changing their parameters. Further, the user can insert an arbitrary transfer function (by setting their poles and zeros) up the fifth order.

The paper is organized as follows. A brief review of the software used to create the app is in Section 2. The app is the described in Section 3 and ideas on how to use it during a lecture are given in Section 4. Finally, conclusions are given in Section 5.

II. CREATION OF THE APP

The software used to create the app has been described in [4] and it is here only briefly reviewed by adding details about the new features. The app is built with Flutter (based on Dart), a single-codebase framework where everything on screen is a widget, covering both visual elements and structural/behavioral aspects like layouts and events. Flutter separates interface from logic using two core widget types: *StatelessWidget*, which is immutable and suits static content, and *StatefulWidget*, which owns a mutable object *State* that triggers rebuilds for interactive or changing user interface. In this app, sliders let users change system parameters (e.g., gain) and observe effects in real time, while text fields allow precise numeric input for repeatable experiments. Each screen includes an interactive *LineChartWidget* that renders the plots, updating dynamically with parameter changes and supporting tap inspection of values (for example at specific times or frequencies). To enhance visualization, the app uses `fl_chart` for smooth, customizable step-response graphs, `syncfusion_flutter_charts` for Bode plots with logarithmic scaling, and `flutter_text` for neatly formatted technical and mathematical text. Then, `syncfusion_flutter_charts` is used to format the pole–zero maps by displaying the poles and zeros with real and imaginary components. Finally, for the Nyquist plot, Syncfusion's `SfCartesianChart` is used to render the points of the frequency response in the complex plane as a scatter plot, featuring dynamic axes and special markers for the critical point $(-1, 0)$. For both the pole–zero map and Nyquist plot, the class dynamically manages the axis limits,

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automatically extending the scale when newly added values exceed the visible range and provides interactive tooltips for detailed data analysis. Together, these choices produce a smooth, customizable, and highly interactive environment for analyzing system dynamics.

III. APP DESCRIPTION

In this section the app is described by highlighting the additional features with respect to the previous version reported in [4].

Through the homepage of the app the user can select which kind of transfer function has to be analyzed: first-order, second-order with complex conjugate poles, second-order with complex conjugate poles and a custom transfer function. In the case of a second-order transfer function, a zero can be subsequently added by using a specific button. Then, the parameters of the transfer function can be defined by writing their values in small boxes and then they can be interactively changed by using sliders. The resulting transfer function is displayed and the step response of the system is plotted. An example with the transfer function

$$G(s) = \frac{1}{(s+1)(10s+1)} \tag{1}$$

(chosen to understand the concept of dominant pole) is shown in Figure 1. Note that the axes can be scaled by suitable buttons. Then, by swiping the screen horizontally, the user can evaluate the pole-zero map (see Figure 6), the asymptotic Bode plot (see Figure 3), and the Nyquist plot (see Figure 4). Alternatively, by clicking an icon, the different plots can be shown one above the other one and the user can visualize them by scrolling the screen. In all the cases, by tapping on a point on the curve, the numerical values corresponding to the point appears on the screen.

If the user selects the 'Custom' option, there is the possibility to insert one or two transfer functions, up to the fifth order, by writing the poles and zeros. This feature can be especially employed by comparing the two transfer functions, for example in order to evaluate how the behavior of a system changes when a zero is moved close to the imaginary axis. An example related to the systems

$$G(s) = \frac{24(s+1)}{(s+1)(s+3)(s+4)} \tag{2}$$

and

$$G(s) = \frac{2.4(s+10)}{(s+1)(s+3)(s+4)} \tag{3}$$

is shown in Figures 5-8, where it appears that the student can easily understand that the zero in -1 causes an overshoot in the step response as well an amplification at low frequencies that is evident in the Bode and Nyquist plots.

IV. USE OF THE APP

From the description of the previous section, it is clear that the app can be used in many contexts.

During the lectures, students can verify by themselves the concepts immediately after they have been explained by the teacher, for example by modifying the parameters (for example, gain, time constant, damping factor, etc.) and understanding their physical meaning. It is also possible for the teacher to assign some task to the students (for example, to evaluate in a high-order transfer function when a pole becomes dominant by moving the other ones far away from the imaginary axis and visualizing that the step response and the frequency response behave like those of a first-order system). Alternatively, the teacher can set quizzes where students can guess the answer using the app and respond using other apps suitable for this purpose. In this case the teacher can also exploit this activity as a feedback of the learning of the students and can, if necessary, improve its explanation.

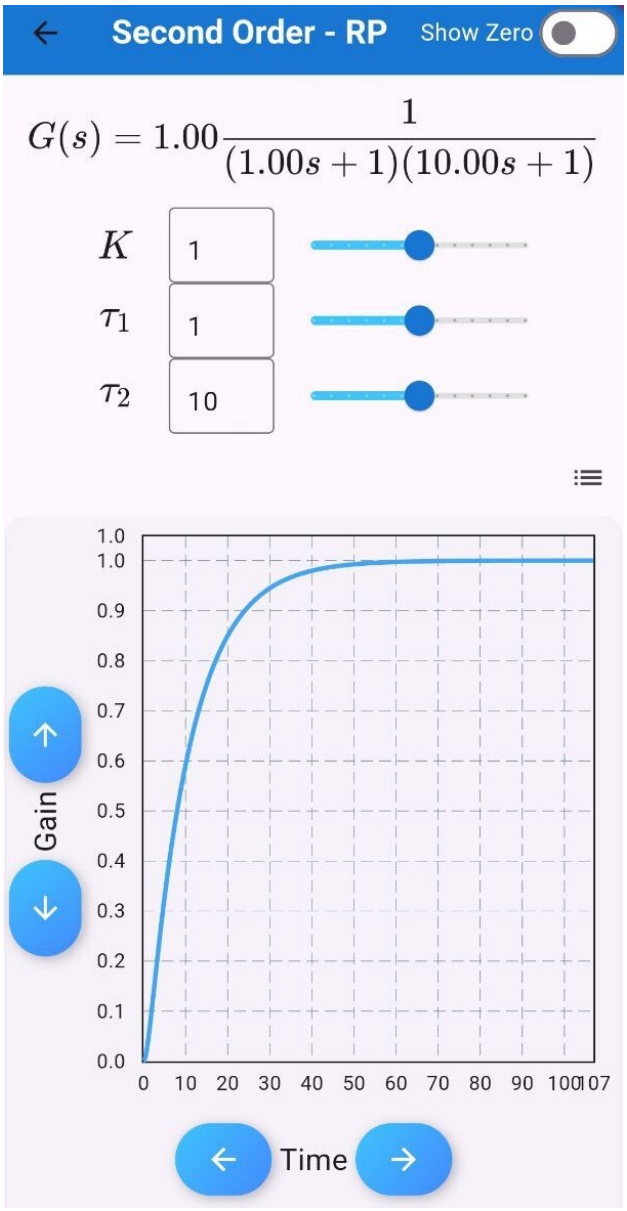


Fig. 1. Screenshot of the step response for a second-order system with real poles.

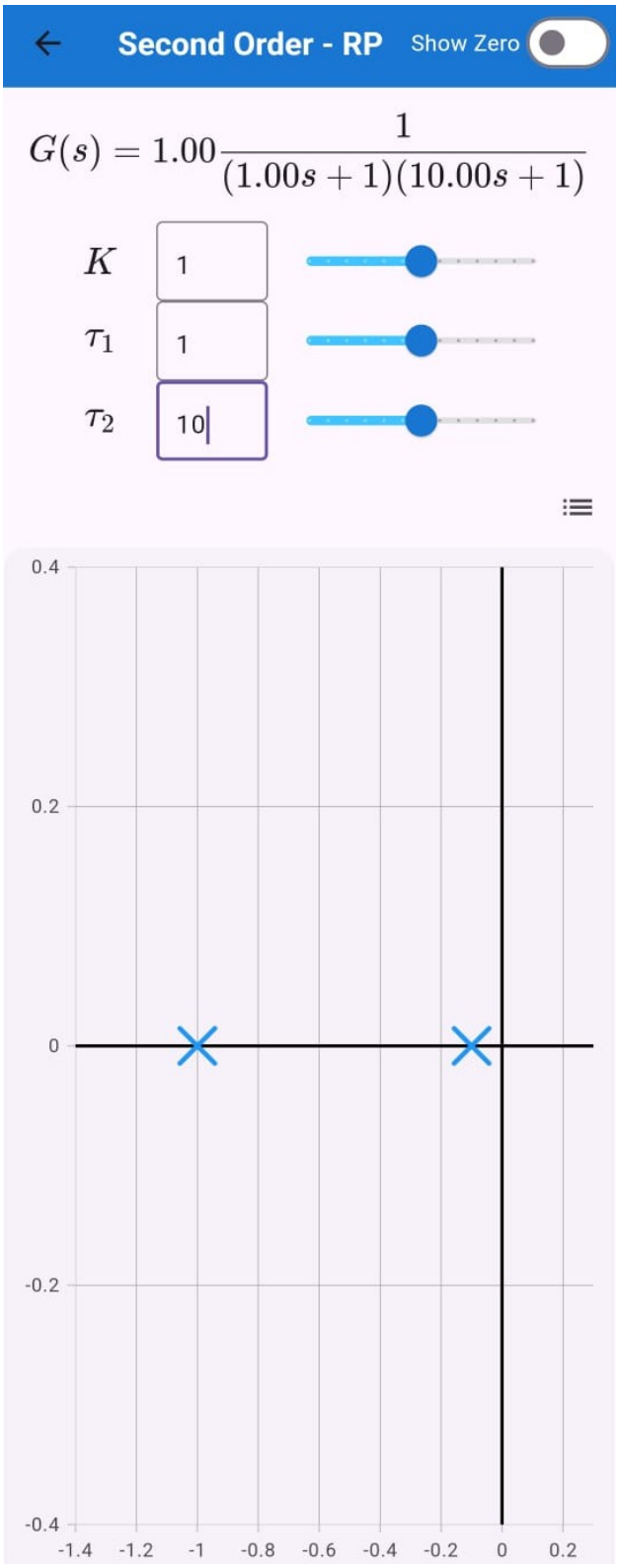


Fig. 2. Screenshot of the pole zero map for a second-order system with real poles.

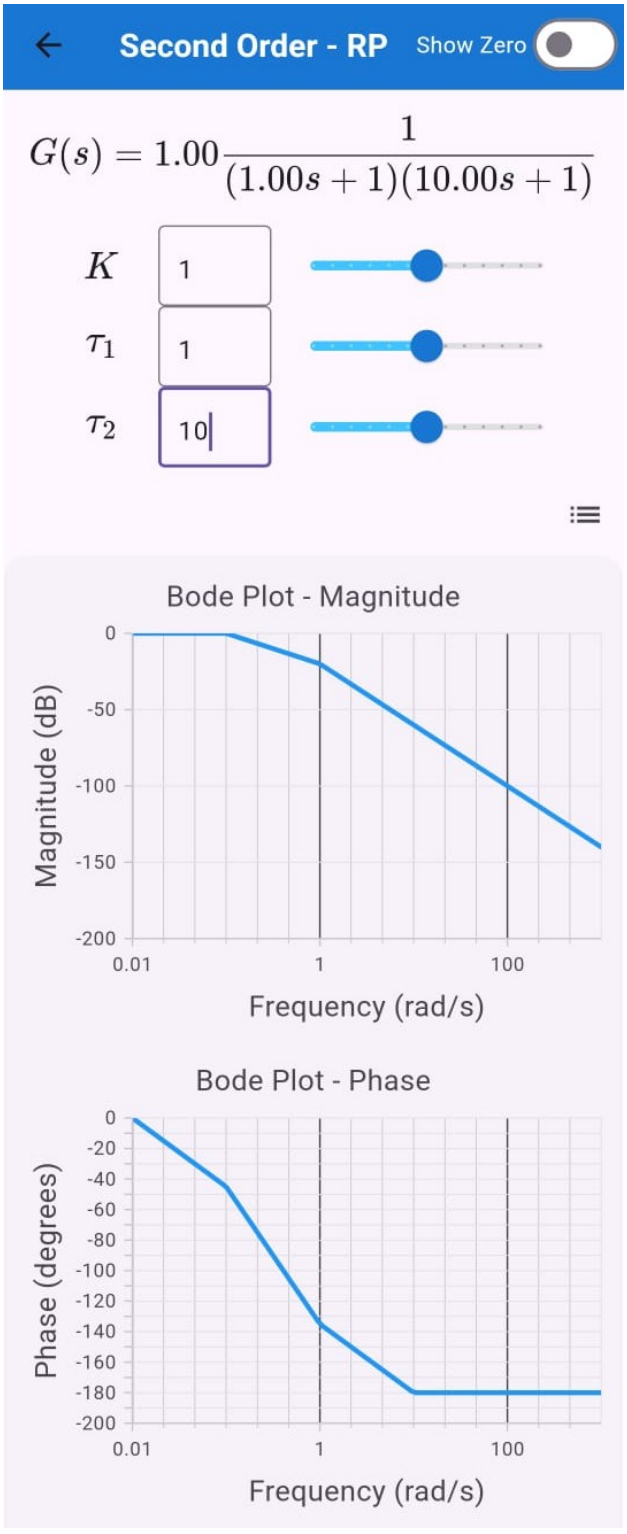


Fig. 3. Screenshot of the Bode plot for a second-order system with real poles.

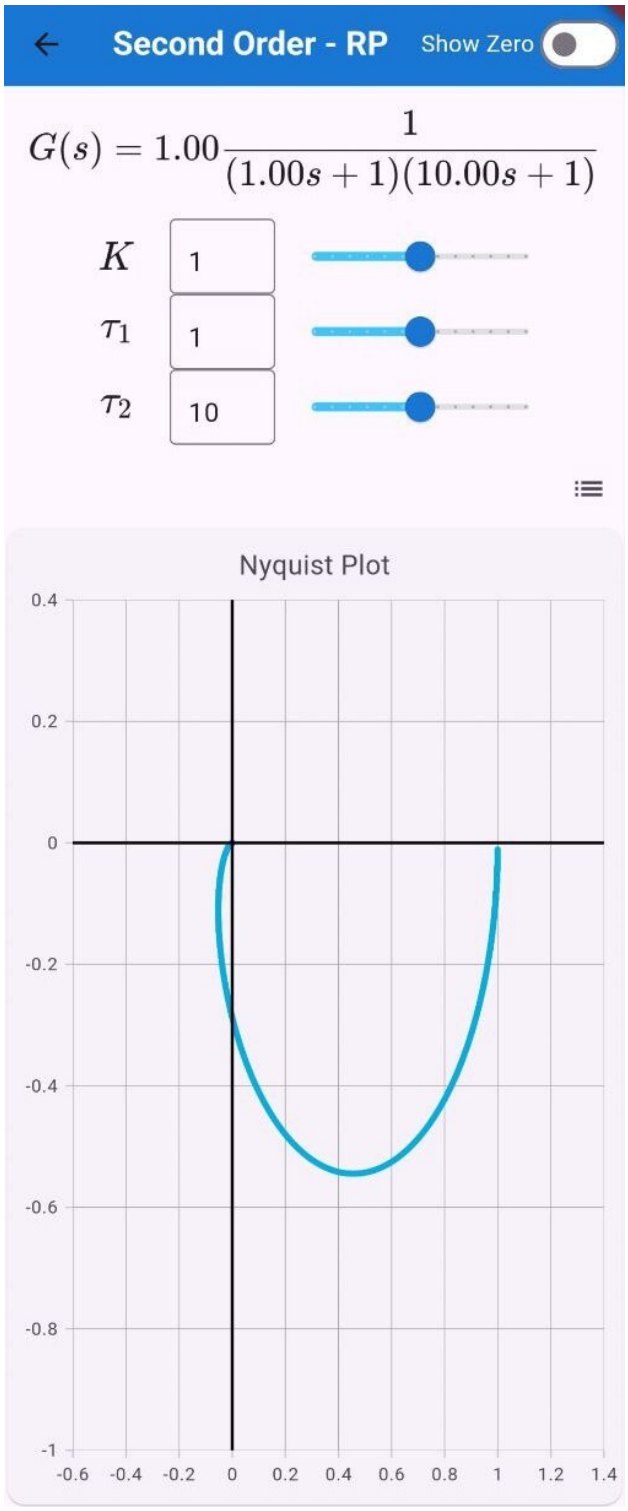


Fig. 4. Screenshot of the Nyquist plot for a second-order system with real poles.

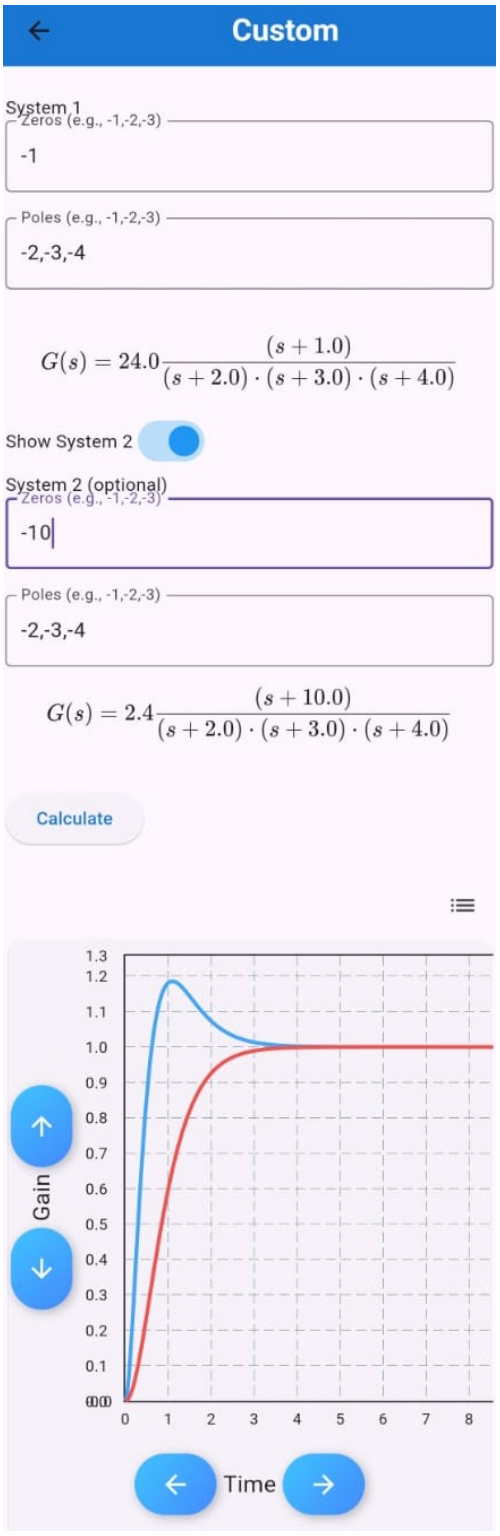


Fig. 5. Screenshot of the step responses for the two systems used for comparison.

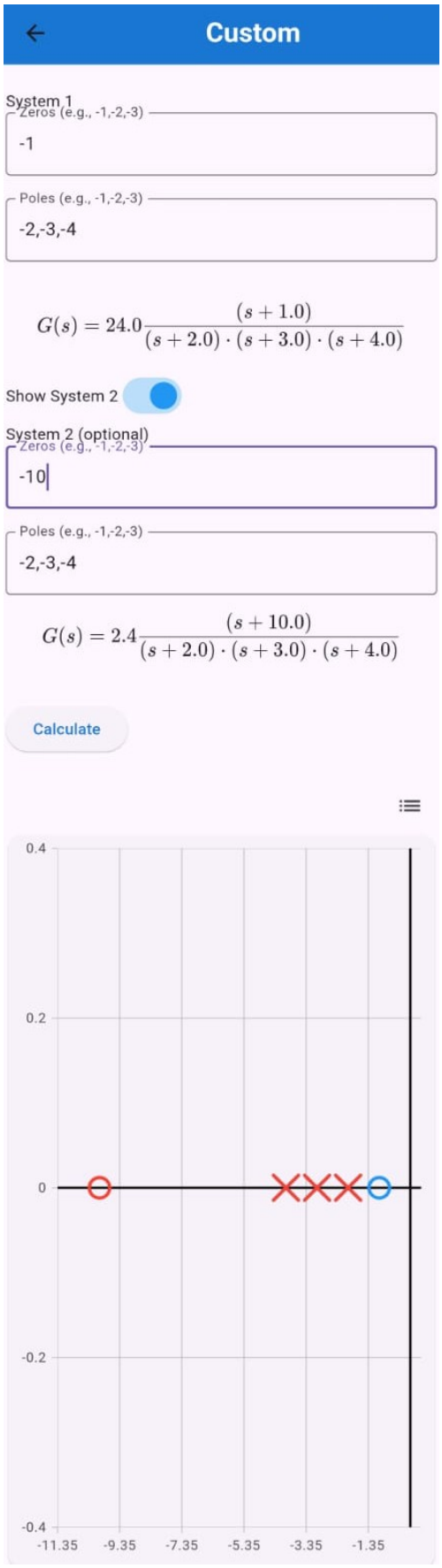


Fig. 6. Screenshot of the pole zero map for the two systems used for comparison.

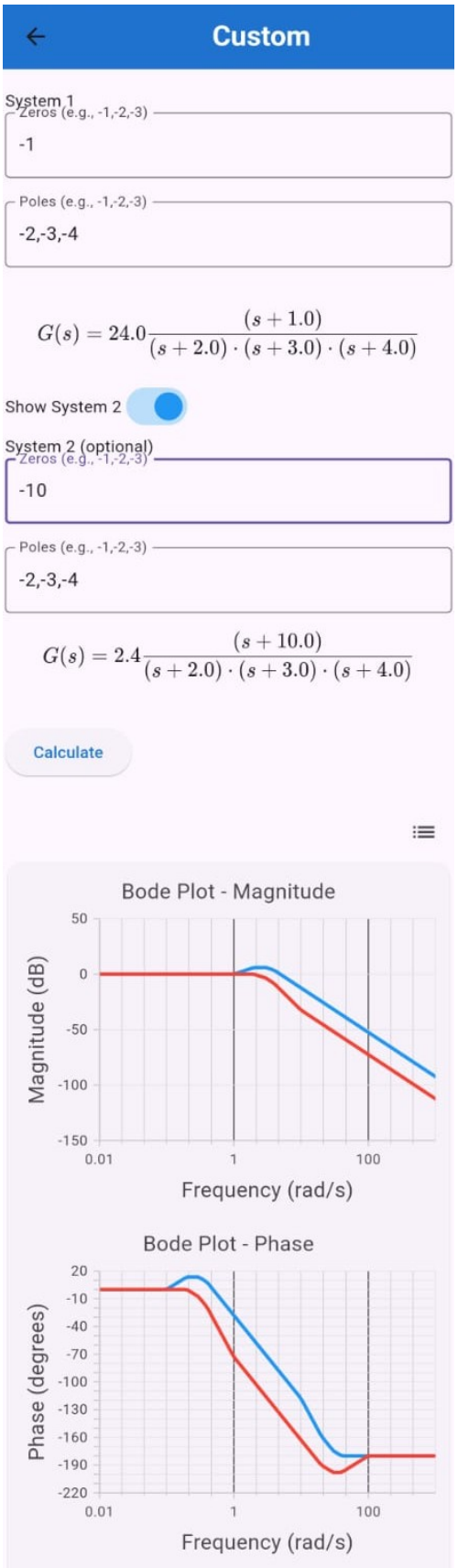


Fig. 7. Screenshot of the Bode plots for the two systems used for comparison.

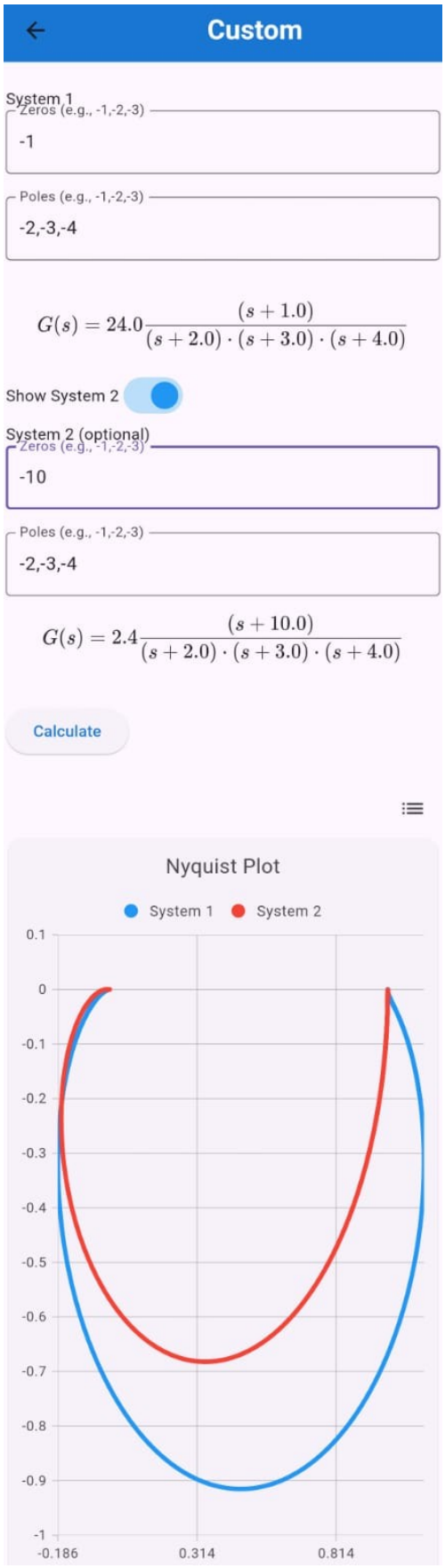


Fig. 8. Screenshot of the Nyquist plot for the two systems used for comparison.

It is however worth underlining that, as the smartphone is a very flexible tool, the possibility to use the app in each part of the day and in each situation makes it very effective in engaging the students.

In order to understand how the app is perceived by the students, its use will be proposed to those attending the course of Fundamentals of Automatic Control held at the University of Brescia (Italy) during the second semester of the academic year 2025-2026. This is the first control course for these students, as they attend the second year of the bachelor program in Industrial Automation Engineering. A survey will be proposed to the students in order to evaluate the app and the results will be given in the final version of this paper if it will be accepted for presentation at the ECC conference.

V. CONCLUSIONS

In this paper we have presented an improved smartphone app to learn concepts associated to transfer functions, In particular, the step response, pole-zero map and Bode and Nyquist plots are shown. Its interactivity allows the user to understand what happens in the different contexts when a parameter is changed so that their physical meaning can be learned easily. The app for Android can be dowloaded by using the QR code of Figure 9.

Fig. 9. QR Code for download the app

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